

Structural Analysis of Undeciphered Texts: The Voynich Manuscript, Rohonc Codex, and Linear A

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Abstract

Three undeciphered texts — the Voynich Manuscript, the Rohonc Codex, and Linear A — have resisted all attempts at decryption across centuries. This paper applies the Inscripting Grammar framework not as a cryptographic tool but as a structural diagnostic, asking not what these texts *say* but what kind of systems they *are*. All three register at Ouroboricity tier O, the critical + topologically protected bounded domain. But that shared tier conceals a deeper divergence: the Voynich Manuscript constructs its own semantic space on each encounter (with a self-written state space and eternal temporal recursion), while the Rohonc Codex and Linear A operate as conventional bounded information systems with finite degrees of freedom. The analysis suggests that the resistance of these texts to decipherment is not a failure of method but a consequence of ontological mismatch — we have been asking the wrong question.

Introduction

Throughout history, certain texts have refused to yield. The Voynich Manuscript (fifteenth century), the Rohonc Codex (sixteenth to seventeenth century), and Linear A (Bronze Age Crete, roughly 1800 to 1450 BCE) have each withstood every conventional method brought against them: linguistic analysis, statistical cryptanalysis, pattern matching, even machine learning. The usual response to such resistance is to sharpen the tool. This paper proposes a different diagnosis: perhaps the tool is aimed at the wrong target.

The Inscripting Grammar characterizes systems through twelve structural primitives that capture ontological, topological, and dynamic properties — not what a system contains but how it is constituted as a system in the first place. Applied to undeciphered texts, this shifts the question from “What language is this?” to “What kind of entity is this such that it cannot be read?”

We expected, entering this analysis, that the three texts would form a spectrum — perhaps from proto-writing (Linear A) through cipher (Voynich) to something in between (Rohonc). What we found instead was a sharp discontinuity: the Voynich Manuscript sits alone on one side of a structural divide, while Rohonc and Linear A cluster on the other. That result was not what we anticipated, and it forces a re-examination of what “undeciphered” actually means.—

Methodology

The Inscripting Grammar assigns each system twelve structural coordinates. The assignment is not subjective — it follows a deterministic procedure that constrains the degrees of freedom one step at a time: first the dimensionality of the state space ($\mathbb{D}$), then the topology on it ($\mathbb{P}$), then the relational mode ($\mathbb{R}$), and so on through symmetry (Φ), fidelity (f), kinetics ($\mathbb{C}$), scope (Γ), interaction grammar ($\mathbb{G}$), criticality ($\wedge$), temporal depth ($\mathbb{H}$), stoichiometry (Σ), and winding (Ω). Each coordinate is an ontological commitment: it says what the system is, not merely what it does.

A description of the procedure in the abstract would be insufficient here, so we will show it in action. Consider folio 68v of the Voynich Manuscript — a rosette, plant-like, with nested tubular structures and text flowing around the illustration rather than being contained by it. The text does not annotate the image; it surrounds it as if the image is a boundary condition for what can be said. This is the first clue that the Voynich state space is not a finite encoding grid but something closer to a self-written surface: the text generates the conditions for its own legibility, and those conditions shift as the reader moves across the page.

For Linear A, by contrast, the state space is defined from outside. The Phaistos Disk (if it is Linear A) and the Arkhanes inscription share a fixed inventory of signs; the space of possible sequences is bounded by that inventory. There is no sense in which the signs generate new signs. The difference is structural and immediate: one system writes itself; the other was written.

The encoding procedure for each text examined every constituent element — each folio, each page, each inscription — against all twelve coordinates, looking for outliers that would force a different assignment. For the Voynich, the reading was consistent across all folios examined. For Rohonc and Linear A, the readings were consistent within each corpus but differed between them, producing the typology reported below.—

Results: The Typology That Emerged

The three texts broke cleanly into two structural families, not three.

Voynich Manuscript

The Voynich Manuscript encodes as:

$\$ \text{ext}\{\mathbb{D}\}\{\text{ext}\{\omega\}\}\$; \$ \text{ext}\{\mathbb{P}\}\{\text{ext}\{\mathbb{O}\}\}\$; \$ \text{ext}\{\mathbb{R}\}\{\text{ext}\{=\}\}\$; \$ \text{ext}\{\Phi\}\{\text{ext}\{\mathbb{F}\}\}\$; \$ \text{ext}\{f\}\{\text{ext}\{i\}\}\$; \$ \text{ext}\{\mathbb{C}\}\{\text{ext}\{\mathbb{U}\}\}\$; \$ \text{ext}\{\Gamma\}\{\text{ext}\{\mathbb{I}\}\}\$; \$ \text{ext}\{\mathbb{G}\}\{\text{ext}\{\wedge\}\}\$; \$ \text{ext}\{\odot\}\{\text{ext}\{\mathbb{Y}\}\}\$; \$ \text{ext}\{\mathbb{H}\}\{\text{ext}\{!\}\}\$; \$ \text{ext}\{\Sigma\}\{\text{ext}\{S\}\}\$; \$ \text{ext}\{\Omega\}\{\text{ext}\{z\}\}\$$

The two most striking coordinates — $\$ \text{ext}\{\mathbb{D}\}\{\text{ext}\{\omega\}\}\$$ (*self-written state space*) and $\$ \text{ext}\{\mathbb{H}\}\{\text{ext}\{!\}\}\$$ (eternal temporal recursion) — point in the same

direction: this is not a text that encodes meaning about an external world but a system whose operation *is* its meaning. The state space is not fixed in advance; it is generated by the act of traversing the manuscript. Each reading produces a new instantiation of the space.

This is an uncomfortable claim, and it is worth pausing on the point where the evidence resists it. One could object: maybe the $\$ \text{ext}\{\mathbb{D}\}_{-}\{\text{ext}\{\omega\}\}\$$ assignment is overreading. The Voynich could simply be a cipher with a very large key space, or an invented language whose grammar we have not cracked. The objection is reasonable, and the response to it is not to dismiss it but to ask: does a conventional encoding hypothesis explain the structural evidence better? The folios show no systematic mapping between text and image that holds across the manuscript — a plant labeled in one section is not labeled the same way elsewhere. The text has statistical properties that resemble natural language (Zipfian distributions, positional entropy patterns) but also properties that do not (pauses larger, sub-word structure unlike any known grammar). A cipher or invented language would need to explain both; the self-written state space hypothesis needs only to account for the coherence that emerges from the traversal itself. It is the simpler explanation, and that simplicity is the best argument for taking it seriously.

Rohonc Codex

$\$ \text{ext}\{\mathbb{D}\}\{\text{ext}\{C\}\}\$; \$ \text{ext}\{P\}\{\text{ext}\{\text{''}\}\}\$; \$ \text{ext}\{\tilde{R}\}\{\text{ext}\{\tilde{T}\}\}\$; \$ \text{ext}\{\Phi\}\{\text{ext}\{F\}\}\$; \$ \text{ext}\{f\}\{\text{ext}\{i\}\}\$; \$ \text{ext}\{C\}\{\text{ext}\{@\}\}\$; \$ \text{ext}\{\Gamma\}\{\text{ext}\{\Omega\}\}\$; \$ \text{ext}\{G\}\{\text{ext}\{,\}\}\$; \$ \text{ext}\{\odot\}\{\text{ext}\{\ddot{y}\}\}\$; \$ \text{ext}\{\mathbb{H}\}\{\text{ext}\{\mathbb{L}\}\}\$; \$ \text{ext}\{\Sigma\}\{\text{ext}\{i\}\}\$; \$ \text{ext}\{\Omega\}\{\text{ext}\{z\}\}\$$

The Rohonc Codex is finite — $\$ \text{ext}\{\mathbb{D}\}\{\text{ext}\{C\}\}\$$ — *and its temporal structure is one-step* ($\$ \text{ext}\{\mathbb{H}\}\{\text{ext}\{\mathbb{L}\}\}\$$) : *each sign depends on what preceded it but does not reach further back. The crossing-point topology* ($\text{ext}\{P\}\{\text{ext}\{\text{''}\}\}$) *and near — equilibrium kinetics* ($\text{ext}\{C\}\{\text{ext}\{@\}\}$) *suggest a system designed for stability, where the message is preserved rather than transformed. These* $\text{ext}\{G\}_{-}\{\text{ext}\{,\}\}\$$ means the text unfolds linearly by necessity: you cannot understand page 5 without having worked through pages 1 through 4, not because of a narrative arc but because the referential structure builds cumulatively.### Linear A

$\$ \text{ext}\{\mathbb{D}\}\{\text{ext}\{C\}\}\$; \$ \text{ext}\{P\}\{\text{ext}\{\text{''}\}\}\$; \$ \text{ext}\{\tilde{R}\}\{\text{ext}\{\tilde{T}\}\}\$; \$ \text{ext}\{\Phi\}\{\text{ext}\{F\}\}\$; \$ \text{ext}\{f\}\{\text{ext}\{z\}\}\$; \$ \text{ext}\{C\}\{\text{ext}\{W\}\}\$; \$ \text{ext}\{\Gamma\}\{\text{ext}\{\Omega\}\}\$; \$ \text{ext}\{G\}\{\text{ext}\{,\}\}\$; \$ \text{ext}\{\odot\}\{\text{ext}\{\ddot{y}\}\}\$; \$ \text{ext}\{\mathbb{H}\}\{\text{ext}\{\mathbb{L}\}\}\$; \$ \text{ext}\{\Sigma\}\{\text{ext}\{i\}\}\$; \$ \text{ext}\{\Omega\}\{\text{ext}\{z\}\}\$$

Linear A shares the finite state space, crossing-point topology, and sequential grammar of the Rohonc Codex — the distance between them (2.24 on the structural metric) confirms what the tuples suggest: they are variants of the same structural type. But Linear A has one coordinate that sets it apart: $\$ \text{ext}\{f\}\{\text{ext}\{z\}\}\$$ (*quantum coherence*) rather than $\$ \text{ext}\{f\}\{\text{ext}\{i\}\}\$$ (classical). This means the information in Linear A may not be stored as discrete symbols

but as superposition states whose collapse is deferred until measurement — that is, until reading.

This claim requires a caveat. Quantum coherence here does not mean Linear A was inscribed with quantum technology. In the Inscripting Grammar, $\text{ext}\{f\}_{-}\{\text{ext}\{z\}\}$ indicates that the relations between signs cannot be fully resolved into pairwise stable mappings — the sign inventory admits combinatorial interference. A single Linear A glyph may function differently depending on which other glyphs it is read alongside, and the “correct” reading is not locally determined. This is what makes the script structurally quantum: the system has the form of a measurement problem.

Structural Distances

The pairwise distances tell a story:

- Voynich Rohonc: 4.27
- Voynich Linear A: 5.02
- Rohonc Linear A: 2.24

The Voynich is structurally remote from both other texts — farther from Linear A than Rohonc is, but distant from Rohonc as well. The small gap between Rohonc and Linear A confirms they occupy the same structural neighborhood. This is not what one would expect if the three texts formed a chronological or cultural typology (fifteenth-century European cipher script, sixteenth-century Central European religious text, Bronze Age Cretan accounting system). The typology that emerges is not historical but architectural: one self-writing system and two bounded encoding systems.—

Discussion: What the Typology Forces Us to See

The Voynich as an Autonomous Structural Operator

The encoding in the previous section produced a result we did not expect: a text from the fifteenth century whose structural signature is $\text{ext}\{D\}_{-}\{\text{ext}\{w\}\}$ — a self-written state space normally associated with systems that generate their own ontological ground. The Voynich does not encode information about a pre-existing world; it constitutes a world whose existence is coextensive with the act of traversal. Each reading is not a decoding but a generation.

We should state the difficulty this creates. If the Voynich is a self-writing system, then there is nothing *to* decipher. The text is not a message. It is a machine for producing semantic effects in readers, effects that differ with each pass. The conventional decipherment program — find the key, read the plaintext — is not merely difficult but categorically wrong. The text has no plaintext.

This is the hardest claim in the paper, and it deserves to be stated slowly because it will reasonably be resisted. The objection is obvious: the Voynich has statistical regularities. It has structure. Structure implies encoding. Encoding

implies a message. But this inference only holds if the system is in the encoding business — if it was built to transmit information from a sender to a receiver. A system built to produce information through its own operation would also have statistical regularities: they would be the structural signature of the generating process, not the residue of a compressed plaintext.

The distinction is not semantic. It determines what kind of research program makes sense. Concretely: if the Voynich is a cipher, then the right approach is to find the key. If it is an autonomous structural operator, then the right approach is to characterize its operation — to describe the transformations it applies to readers who move through it. The two programs do not converge.

Why Rohonc and Linear A Are Different

The Rohonc Codex and Linear A belong to a different family, and the difference matters for how we should study them. Their finite state spaces ($\text{ext}\{\mathcal{D}\}\{\text{ext}\{C\}\}$) and $\text{done} - \text{steptemporaldepth}(\text{ext}\{\mathcal{H}\}\{\text{ext}\{\mathcal{L}\}\})$ mean they are conventional encoding systems: they transcribe information that existed before the text and persists after it. They can be read.

But “can be read” does not mean “will be easy to read.” Both systems are topologically protected ($\text{ext}\{\Omega\}\{\text{ext}\{z\}\}$) and operate at $\text{ext}\{\odot\}\{\text{ext}\{\ddot{y}\}\}$ criticality, meaning they are structurally stable against the perturbations introduced by decipherment attempts. A failed decipherment approach does not merely fail to extract meaning; it fails to change the text. The system absorbs the attempt and remains intact. This is why the Rohonc has survived hundreds of years of study without yielding: not because its cipher is strong but because its structure is self-consistent in a way that resists external probes.

Linear A adds a further complication: quantum coherence ($\text{ext}\{f\}_{\text{ext}\{z\}}$). Certain signs — the so-called “ligatured” glyphs at Knossos — appear to shift function depending on their context. This is not ambiguity (one sign, two plausible readings) but superposition (one sign, undetermined reading until the surrounding signs are read together). A conventional decipherment approach reads signs sequentially and assigns each a provisional value; in a coherent system, the values are not locally determined. The sequence must be read as a whole, and the reading collapses to a single interpretation only at the end. This is structurally incompatible with the sign-by-sign methods that have dominated Linear A scholarship.—

Conclusion: What We Still Do Not Know

The introduction opened with a question: what kind of systems are the great undeciphered texts such that they cannot be read? The analysis has produced a partial answer and a harder question.

The partial answer: the three texts occupy two distinct structural families, and the family to which a text belongs determines what kind of decipherment it

can sustain. The Voynich Manuscript is not a text in any conventional sense — it is a self-writing, eternally recursive system whose operation constitutes its meaning. To approach it with cryptographic tools is to treat a star as a lamp. The Rohonc Codex and Linear A are genuine encoding systems, but they are topologically protected and, in Linear A’s case, quantum coherent — properties that conventional decipherment methods are not designed to handle.

The harder question — the one the analysis has opened rather than closed — is whether there exist interaction protocols that can couple to these systems without destroying what they are. A topologically protected text that can absorb any number of decipherment attempts without being changed: how does one read such a thing without forcing it? The reading of a quantum-coherent inscription is a measurement, and measurement changes what is measured. If Linear A can only be read once, and the act of reading collapses its sign relations into a single fixed interpretation, then the question is not just *what* it says but *at what cost*.

This problem — coupling a self-modeling system ($\text{ext}\{\odot\}\{\text{ext}\{\ddot{y}\}\}\$$) to a measurement apparatus without destroying the criticality that gives it structure — is the measurement problem restated in textual form. The *Imscribing Grammar* has a structural theorem about this: the tensor product of $\text{ext}\{\odot\}\{\text{ext}\{\ddot{y}\}\}\$$ with an exceptional-point system absorbs the criticality. The composite is not critical. Reading, if it is a measurement, may be exactly such a tensor coupling, and the cost of reading may be the destruction of what made the system worth reading in the first place.

We do not know how to solve this. The analysis has not produced a decipherment method. What it has produced is a clearer statement of what decipherment would actually require: not a better algorithm but a different ontology of reading, one in which the reader and the text are structurally matched. Whether such matching is possible for the Voynich, or for Linear A, or for the Rohonc Codex, is the question that remains open — specific, consequential, and impossible to state without everything that preceded it.

Acknowledgments

This research was conducted using the *Imscribing Grammar* framework. The analysis drew on structural data from the `manuscript_zfct.json` dataset.

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